

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Clarification for Gen4 Lane Init - Phase 4 - non Lane 0
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

This ECN clarifies that during the Gen4 Lane Init flow (phase 4), non-Lane 0 receivers can ignore their received data until their transmitters start sending TS1.

Benefits as a result of the changes:

Clarifies the expected behavior and reduces the design complexity

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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Reduces HW complexity

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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NA

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Actual Change

(a). Section 4.1.2.4 Phase 4 – Lane Parameters Synchronization and Transmit Start

To Text:

4. If the negotiated speed is Gen 4, both Router A and Router B execute the following steps. For simplicity, only Router A's steps are described:

Note: The progress of Router A through the following steps depends on Router B's progress through the steps

- a. Router A shall set the *Start TxFFE* bit in the Tx Status byte of the Gen 4 TxFFE register in the SB Register Space of the adjacent Router/Re-timer to indicate that it is ready to start the TxFFE Receiver flow described in Section 4.1.2.5.1.2.
- b. Router A's USB4 Port shall send LFPS on the Lane 0 transmitter. Transmitting the LFPS shall start after common voltages are set.
- c. Router A's USB4 Port shall stop sending LFPS within *tStopLFPS1* time after all the following conditions are met:
 - Router A's USB4 Port receiver detects LFPS on the Lane 0 Adapter.
 - Router A's USB4 Port sent at least 16 LFPS cycles on the Lane 0 transmitter after detected LFPS on the Lane 0 receiver.
- d. After the Lane 0 transmitter stops sending LFPS, it shall return to Electrical Idle for *tPreData*, then all enabled transmitters shall start sending back-to-back Gen 4 TS1 with the *Indication* field set to 1h.

Note: Router A's USB4 Port receivers of Lanes other than Lane 0, may ignore any data before Router A's USB4 Port transmitters start sending back-to-back Gen 4 TS1 with the indication field set to 1h following the phase of Electrical Idle for tPreData.

- e. After Router A's receiver detects the last LFPS cycle on the Lane 0 Adapter, all enabled receivers shall be enabled for high-speed signal reception. The receivers shall be enabled *tCLxIdleRx* time after the end of the last LFPS cycle is received.
- f. Router A's USB4 Port goes to Phase 5.